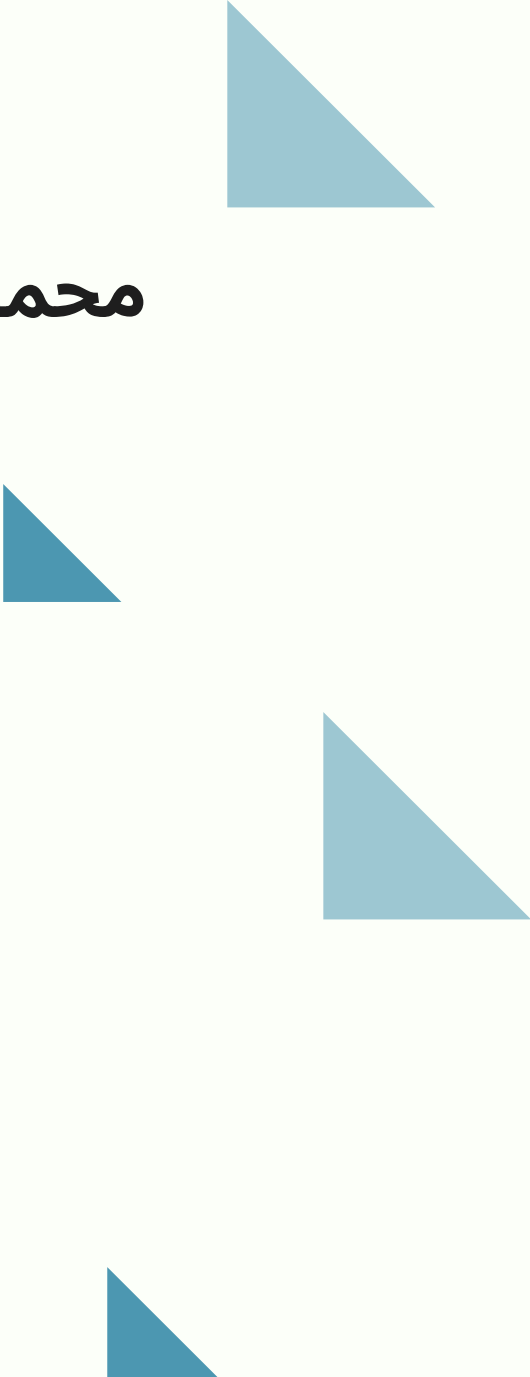




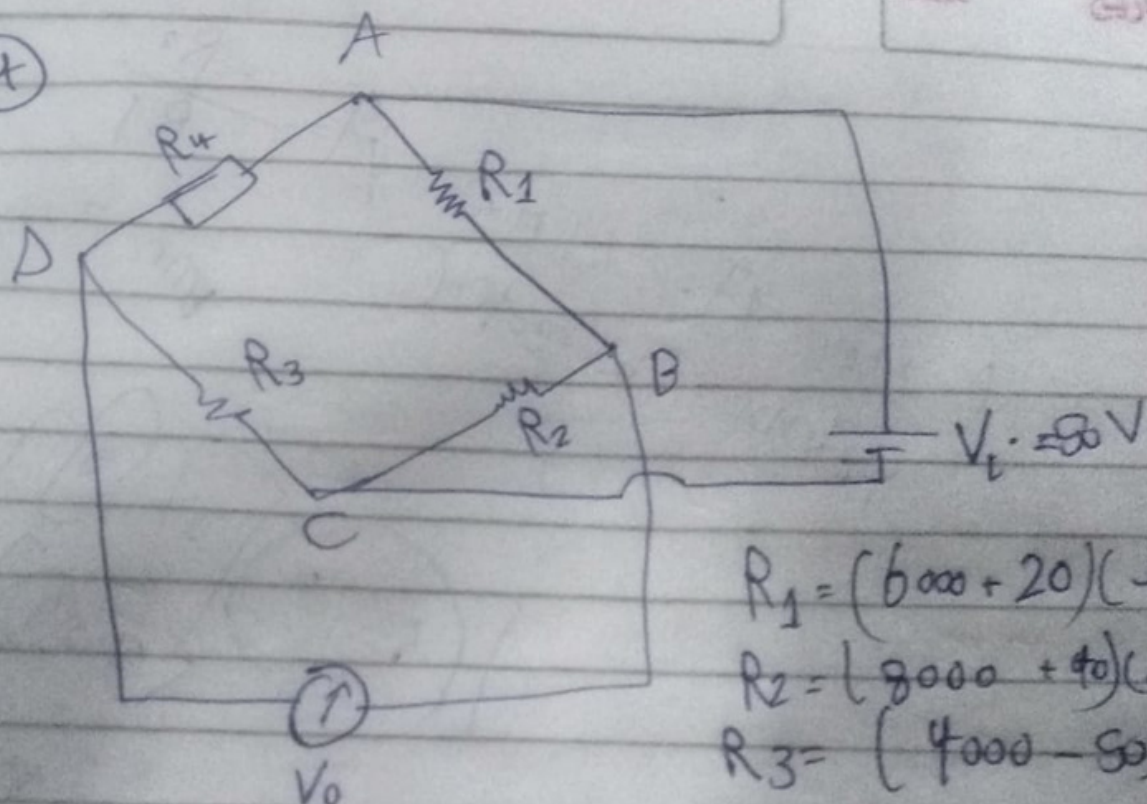
Bio-measurements Assignment 3

محمود بهاء الدين لطفي

Sec: 2



(4)



$$R_1 = (6000 + 20) (\Omega)$$

$$R_2 = (8000 + 40) (\Omega)$$

$$R_3 = (4000 - 50) (\Omega)$$

$$R_4 = (3000 - 30) \Omega$$

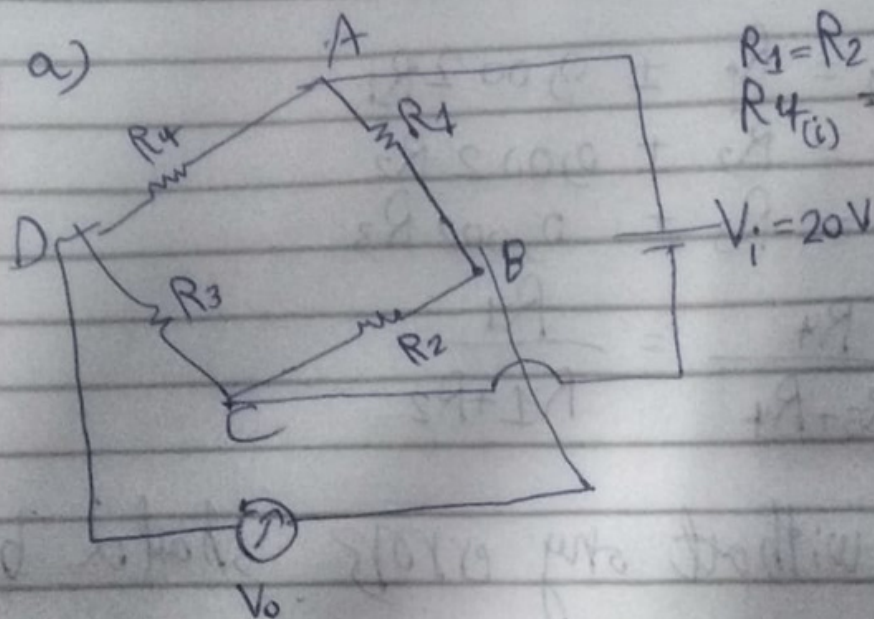
$$V_o = V_C - V_D = V_{AB} - V_{AD} = I_1 R_1 - I_2 R_2$$

$$\therefore V_o = \frac{V_i R_1}{R_1 + R_2} - \frac{V_i R_4}{R_3 + R_4}$$

$$= 50 \left[\frac{6000 + 20}{6000 + 20 + 8000 + 40} \right] - \left[\frac{3000 - 30}{3000 - 30 + 4000 - 50} \right]$$

$$= -0.05128 \text{ (V)}$$

⑥ a)



$$R_1 = R_2 = R_3 = 500 \Omega$$

$$R_{4(i)} = 500 \Omega$$

to get the R_4 assume $(2N)$ applied on the strain gauge:-

$$\therefore R_4 = 500 + 2(\text{sensitivity}) = 500 + 2(0.5) = 501 \Omega$$

$$\therefore V_0 = \frac{V_i R_4}{R_3 + R_4} - \frac{V_i R_1}{R_1 + R_2} = 20 \times \left(\frac{501}{1000} - \frac{500}{1000} \right)$$

$$= 10^{-2} (V) = 10 (mV)$$

then the sensitivity = $\frac{10}{2} = 5 (mV/N)$

b) at $F = 500 N$ $\therefore R_4 = 500 (1 + 0.5) = 750 \Omega$

$$\therefore V_0 = 20 \left(\frac{750}{1250} - \frac{500}{1000} \right) = 2 (V)$$

$$\textcircled{8} \quad R_1 = R_1 \pm 0.002 R_1$$

$$R_2 = R_2 \pm 0.002 R_2$$

$$R_3 = R_3 \pm 0.002 R_3$$

$$\therefore \frac{R_4}{R_3 + R_4} = \frac{R_1}{R_1 + R_2}$$

$\therefore R_4$ without any errors should be:

$$R_4 = \frac{R_1 R_3}{R_2} = \frac{(1 + 0.002)^2 R_1 R_3}{1 - 0.002 R_2} = 1.006 \frac{R_1 R_3}{R_2}$$

$$\therefore \text{error} = \pm 0.6\%$$